「Def. Let G be a Group. Define the upper central series $\{Z_n(G)\}_{n=0}^{\infty}$ as:

$Z_0(G) = 1$,

$Z_1(G) = Z(G)$,

$\vdots$

$Z_{i+1}(G) \overset{\text{def}}{=} \varphi_i^{-1}[Z(G/Z_i(G))]$ where $\varphi_i : G \to G/Z_i(G) : x \mapsto xZ_i(G)$

Note: $1 = Z_0(G) \le Z_1(G) \le Z_2(G) \le \cdots$

because $Z_i(G) \le \varphi_i^{-1}[Z(G/Z_i(G))]$ given by

$$x \in Z_i(G) \Rightarrow \varphi_i(x) = xZ_i(G) = 1 \cdot Z_i(G) \in Z(G/Z_i(G))$$

$$\Rightarrow x \in \varphi_i^{-1}[\varphi_i(x)] \subseteq \varphi_i^{-1}[Z(G/Z_i(G))]$$

and, preimage of Normal is normal, the definition is well-defined.

Clearly, $Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G))$

$$Z_{i+1}(G)/Z_i(G) = \varphi_i^{-1}[Z(G/Z_i(G))]/Z_i(G) \quad (\varphi_i \text{ onto})$$

$$= \varphi_i[\varphi_i^{-1}[Z(G/Z_i(G))]] = Z(G/Z_i(G))$$

「Def. A Group G is called Nilpotent if: for some $c \ge 0$, $Z_c(G) = G$.
The smallest such $c \ge 0$ is called the nilpotent class of G.

「Prop. Let P be a group of order p^a, where p is prime and $a \ge 1$.
Then P is Nilpotent of nilpotent class at most $a-1$.

Proof. Since P-group has no trivial center, for all $i \ge 0$,

$$|P/Z_i(P)| > 1 \Rightarrow Z(P/Z_i(P)) \neq 1$$

Therefore, $Z_i(P) \neq P \Rightarrow |P/Z_i(P)| > 1 \Rightarrow Z(P/Z_i(P)) \neq 1 \Leftrightarrow Z_{i+1}(P)/Z_i(P) > 1$

This implies $|Z_{i+1}(P)| \ge p \cdot |Z_i(P)|$, further, $|Z_{i+1}(P)| \ge p^{i+1}$

In particular, $|Z_a(P)| \ge p^a$, at least $i = a$, $Z_a(P) = P$.

If nilpotent class of P is exactly a, then for all $i \ge 0$, $|Z_i(P)| = p^i$.

Then, $P/Z_{a-2}(P)$ has order p^2, so is abelian. Now,

$$Z_{a-1}(P) = \varphi_{a-2}^{-1}[Z(P/Z_{a-2}(P))] = P$$

which is Contradiction.

Upper, lower, Central series.